

SESSION 1 · SCALING MEDIA SERVICES · SPEAKER SLIDES

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9 ways MOQ changes Media Delivery

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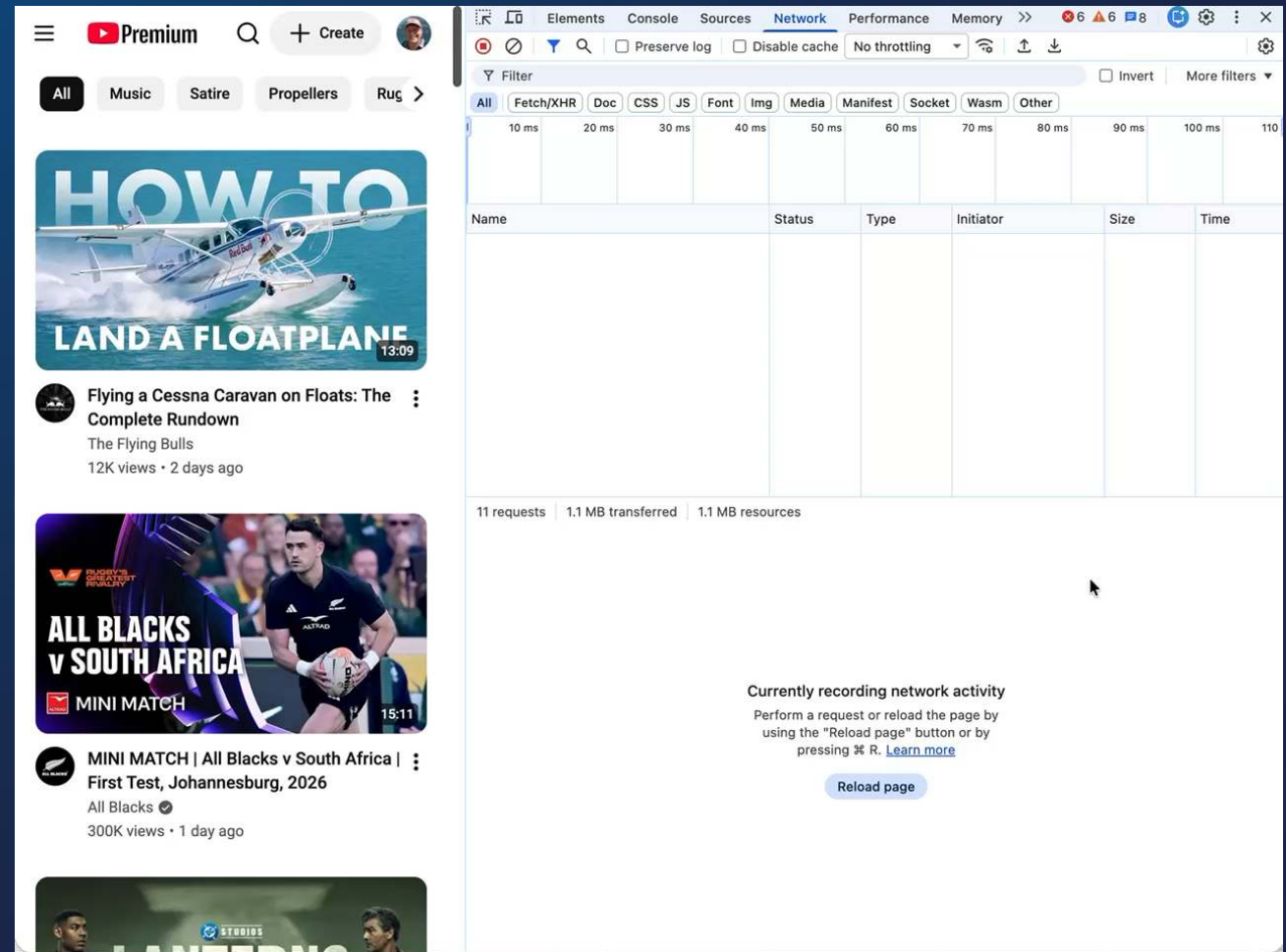
Sept 2026

1. MOQ isn't only for media

The core MOQ Transport spec is intentionally **payload agnostic**.

MOQT can move media well, but it can also *move all the other data associated with modern media playback*:

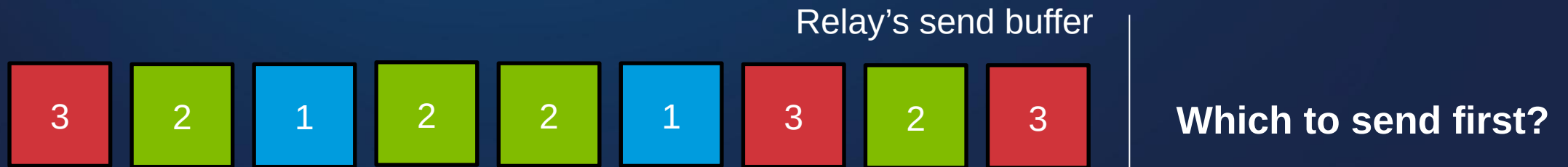
- Manifests
- Sports scores
- User interactivity
- Token refresh instructions
- QoE Data
- Log files
- Advertising decisioning



2. Priorities between flows in the same connections

MOQT allows **relative priorities** to exist between streams and datagrams, and to be defined separately both publishers and subscribers.

Business impact: prioritize audio over video, or base layer SVC over enhancement layers.



Publishers ranks:

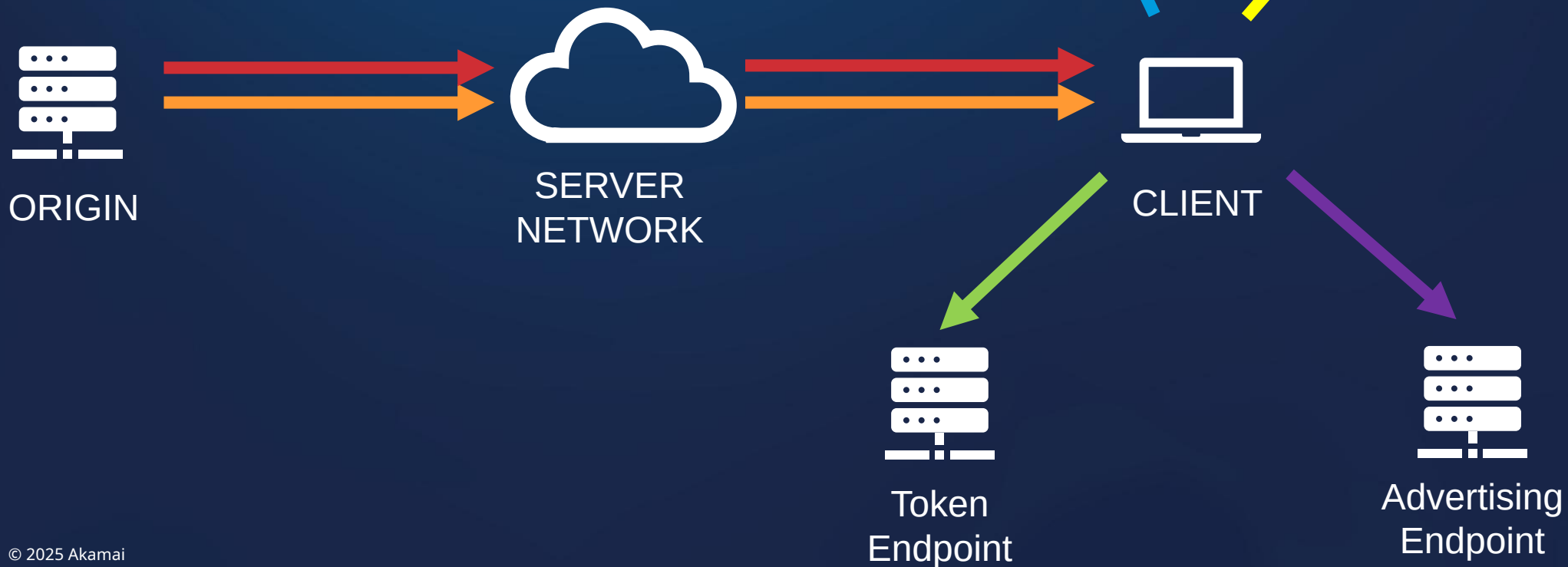
Blue - highest priority
Green - medium priority
Red - low priority

Subscriber ranks:

Green - highest priority
Blue - medium priority
Red - low priority

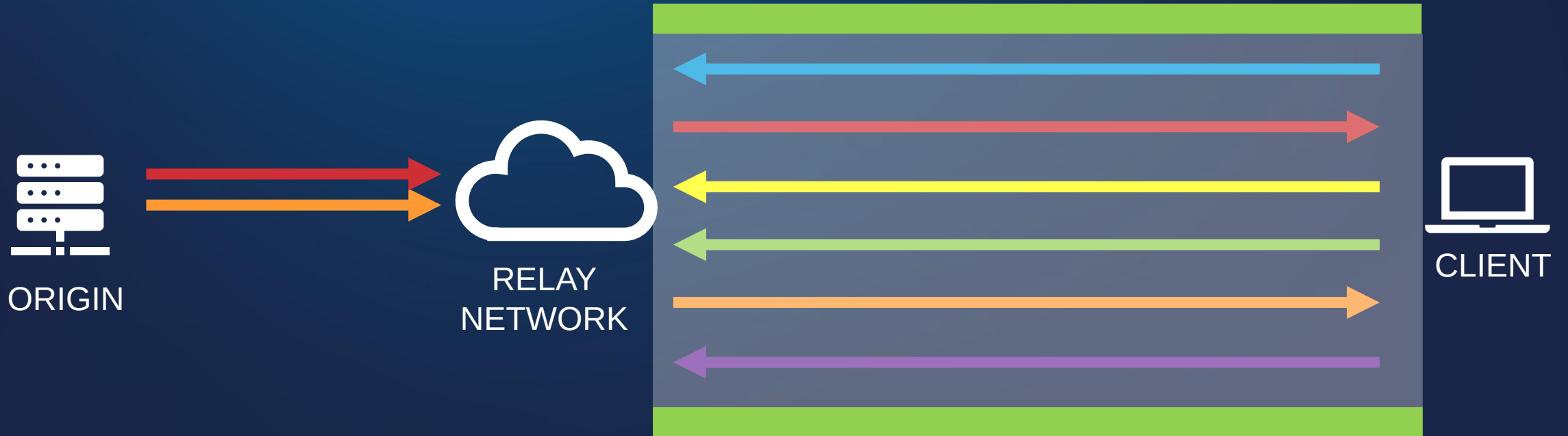
5. MOQT is bidirectional

Information flows under HTTP delivery



5. MOQT is bidirectional

Information flows under MOQT



Single QUIC TLS 1.3 connection

3. Network filtering

With MOQT, the distribution network can **intelligently filter** the incoming data using **Object Filters**

Use cases:

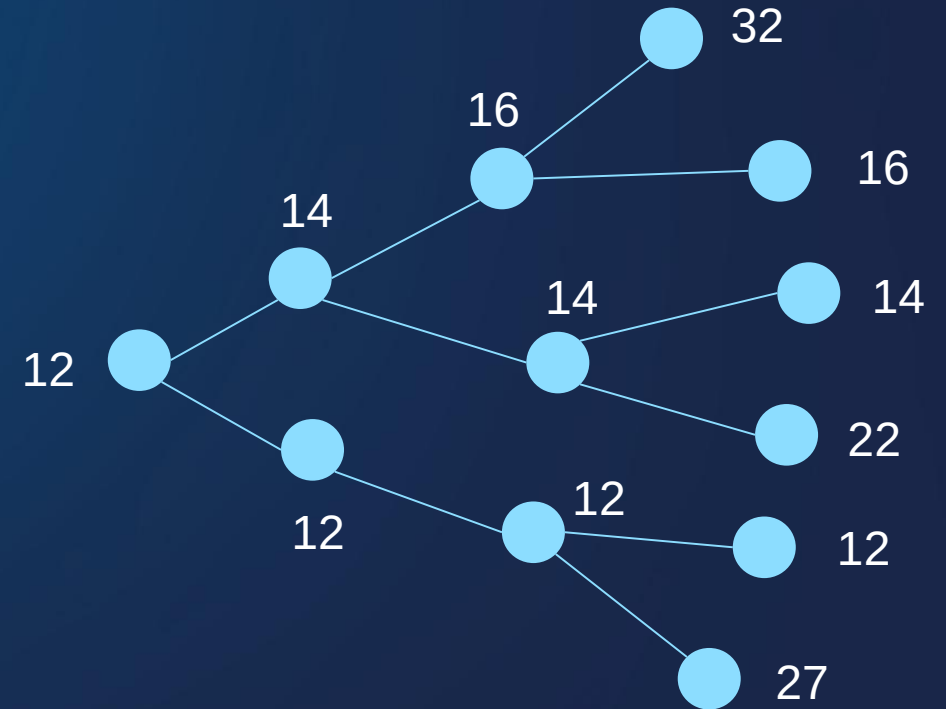
Meetings 100 people online, see/hear last 4 active speakers

Security Cams 100 cams, monitor top 6 with most activity

Viral Videos 100 live video feeds, see top 4 most popular

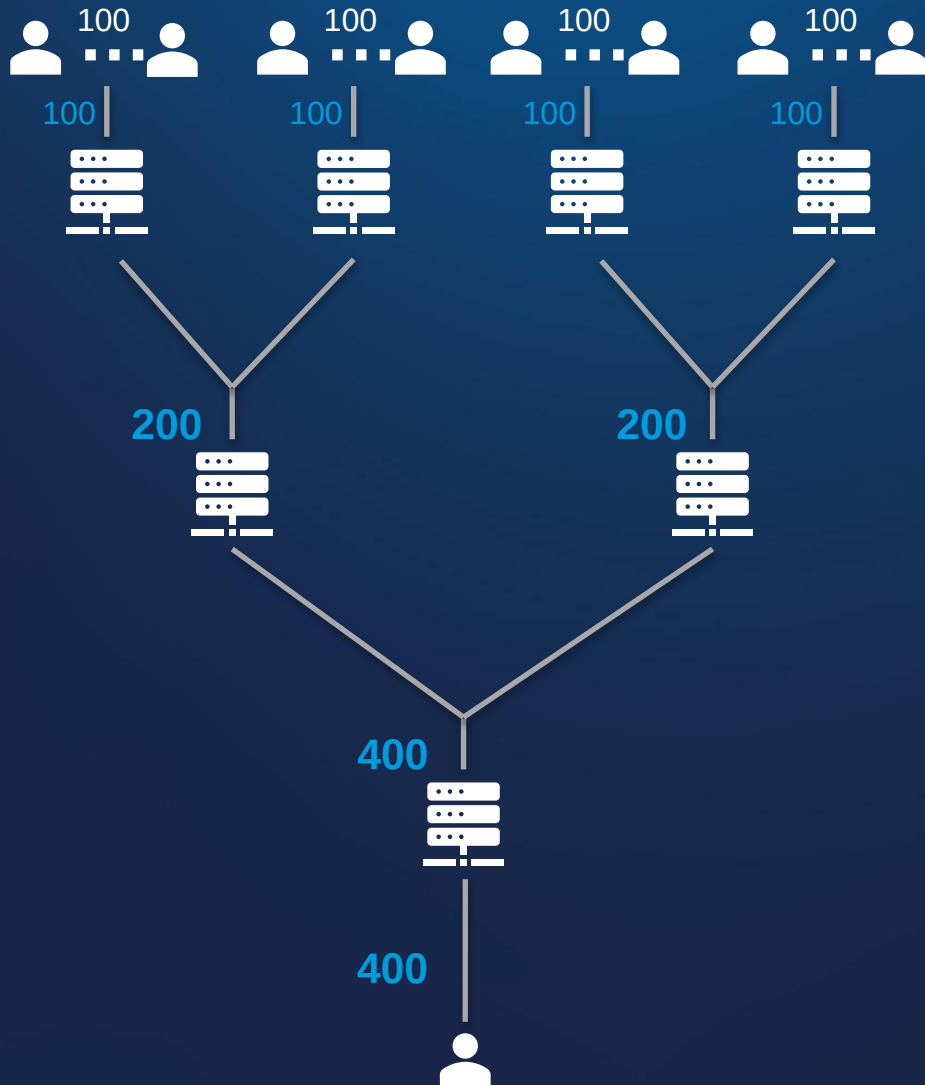
Sensors, Telemetry 100 temperature sensors, get top 4 hottest

Chat Tracks show me the last 25 comments from 10,000 chat clients.

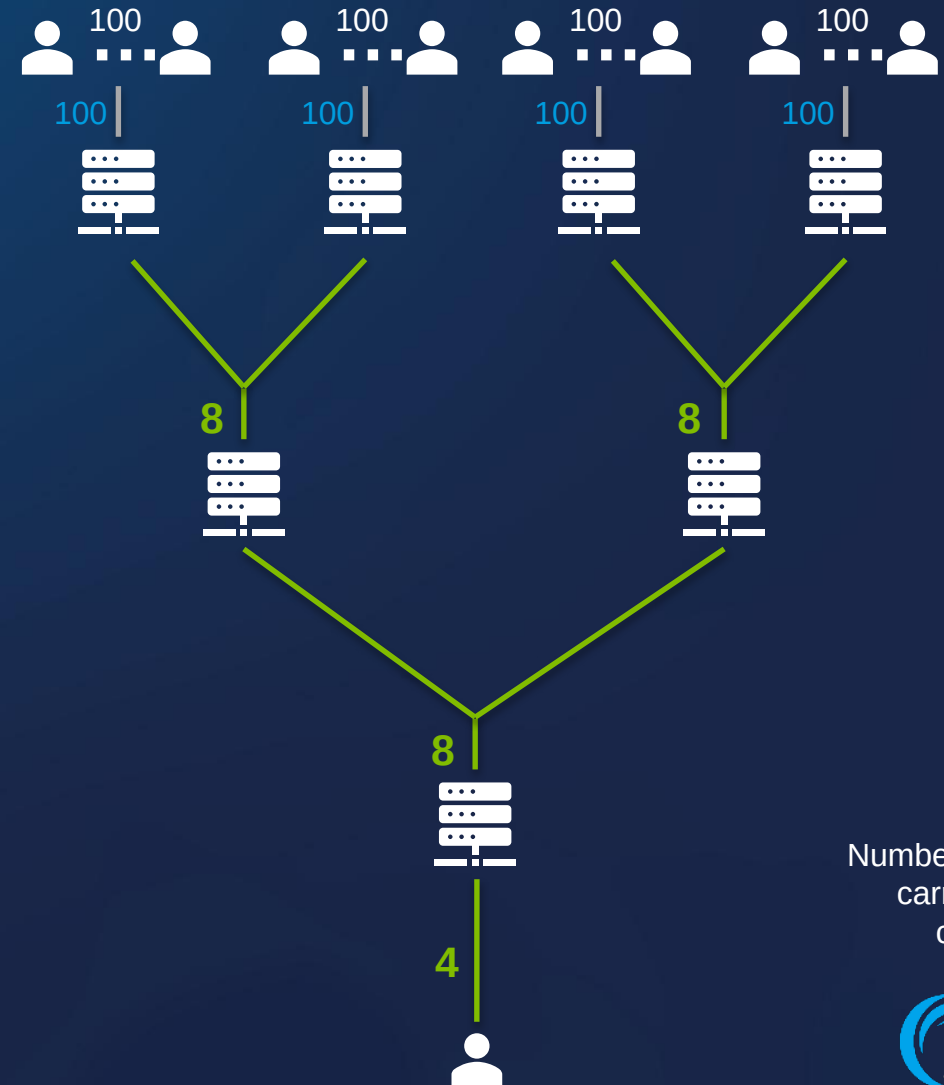


CDN midgress traffic for fan-in with & without top-N filtering

Without top-N in place at each tier

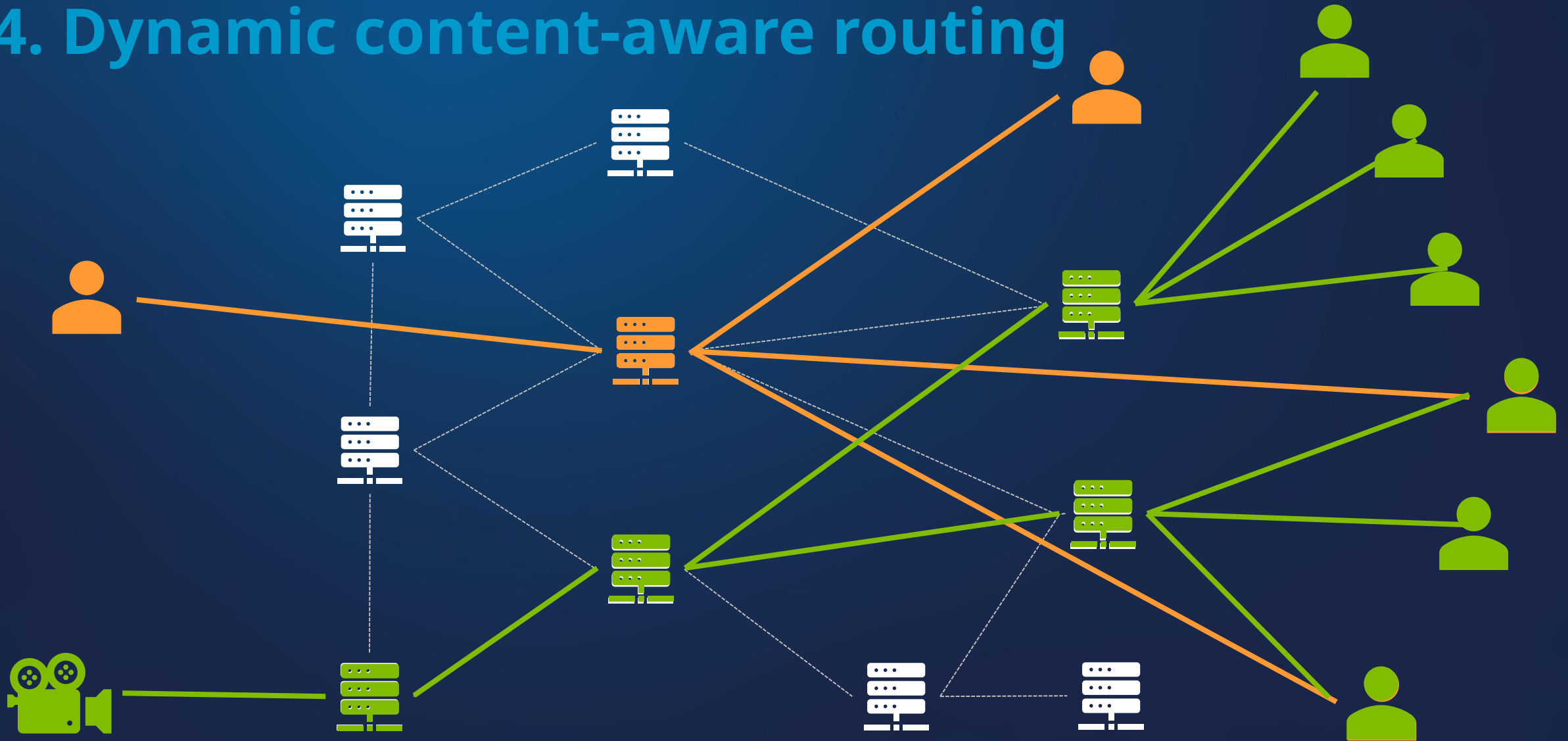


With top-4 in place at each tier



Number of tracks
carried by the
connection

4. Dynamic content-aware routing



Shortest path for real-time media

6. Edge prewarming via SUBSCRIBE forward parameter

A Subscription can request a publisher to not forward Objects for a given track by setting the Forward field to 0.

Use SUBSCRIBE_UPDATE to change the forwarding state.

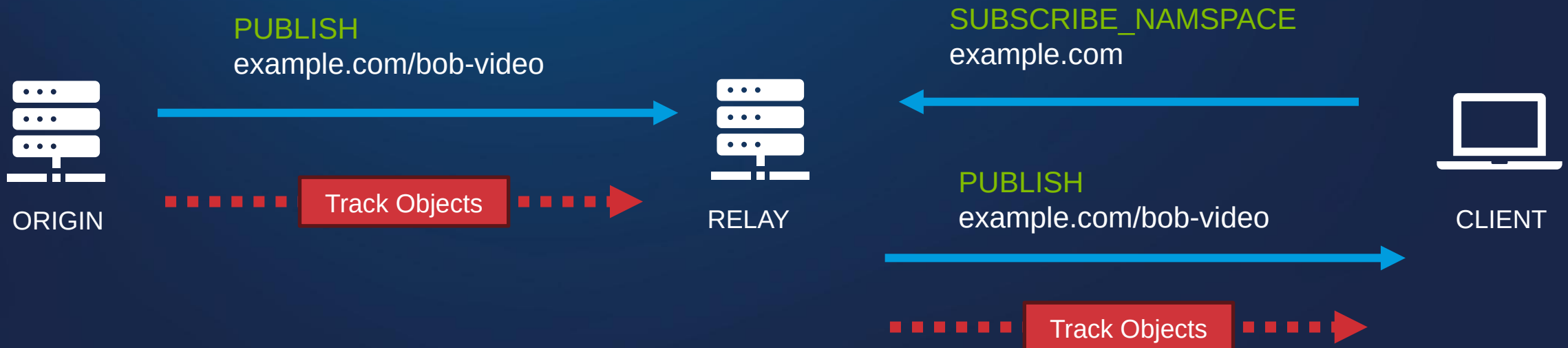


Useful for quick ABR switching as alternate tracks are pre-warmed at the edge.

Also a DDOS vector so relays may refuse the request.

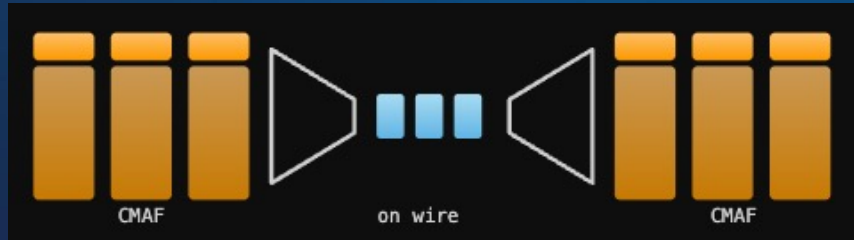
7. SUBSCRIBE to things you don't yet know

Allows receipt of all tracks produced under a namespace, without having to know their names. Useful if the track names are generated on the-fly, for example, people joining a web conference



8. Better CMAF - LOCMAF (low overhead CMAF for MOQ)

Hugo Björs (MSc thesis) · Torbjörn Einarsson · Eyevinn · 2026



- A new, compact wire format for streaming CMAF over MOQ Transport
- A typical CMAF chunk header for one sample is around ~104 bytes.
- LOCMAF observes that consecutive moof boxes are almost identical and delta-encodes them.



<https://locmaf.dev/>



track	sample	CMAF	LOCMAF	saved
audio_128kbps_aac	128.0 kbps	171.5 kbps	131.9 kbps	23.1 %
video_400kbps_avc	373.2 kbps	396.4 kbps	376.5 kbps	5.0 %

9. Shared software development OpenMOQ



- A software consortium founded to develop high performance, open-source MOQ software.
- Founded January 2026 – now have 20 members
- Get the software at <https://github.com/openmoq>



CDN77



ORACLE



Synamedia



YouTube



Fishjam



MAINSTREAMING®



nano
COSMOS



VINDRAL



WOWZA



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KLAGENFURT



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Thank you for your time.